SRM INSTITUTE OF SCIENCE AND TECHNOLOGY · COLLEGE OF ENGINEERING AND TECHNOLOGY
Department of Electronics and Communication Engineering · Academic Year 2026–2027 (ODD)

LLT1 Simulation: Electromagnetic Theory and Transmission Lines

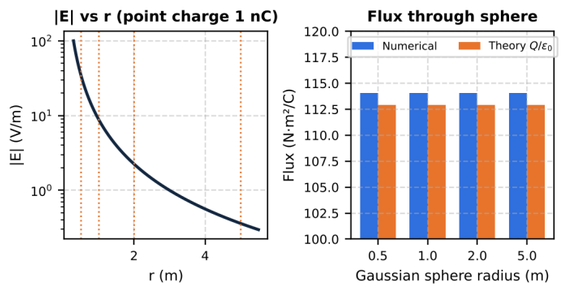
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21ECC205T – Electromagnetic Theory and Interference | Year II / Sem III / ECE | Tool: Python (NumPy, Matplotlib)

1 Gauss's Law: Flux through a Sphere — CO1 · BL4

Aim: Compute the electric flux of a 1 nC point charge through spherical Gaussian surfaces of radius 0.5, 1, 2 and 5 m by numerical surface integration, and compare it with Q/ε_0.

Model: $E = Q/(4\pi\varepsilon_0 r^2)\,\hat{r}$; $\Phi = \oint E \cdot dA$, summed over a 100 × 100 (θ, φ) grid with $dA = r^2 \sin\theta\, d\theta\, d\varphi\, \hat{r}$.



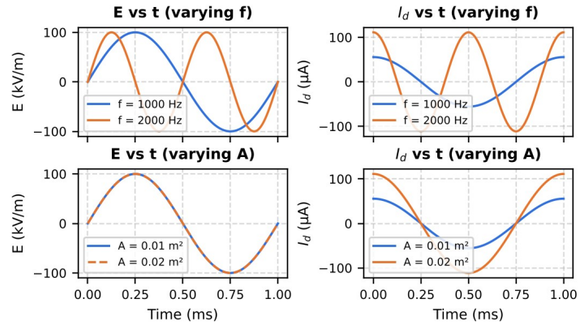
Result: Theoretical Φ = Q/ε₀ = 112.94 N·m²/C. Numerical Φ = 114.07 N·m²/C for every radius (|error| = 1.13, i.e. 1.00 %).

Inference: The flux is the same for all radii: it depends only on the enclosed charge, not on the size of the Gaussian surface (Gauss's law). The ~1 % error comes from the discretised (θ, φ) grid and falls with a finer mesh.

2 Displacement Current in a Capacitor — CO2 · BL4

Aim: Simulate a time-varying field E(t) = E_0 sin ωt between the plates of a parallel-plate capacitor and the resulting displacement current; study the effect of frequency and plate area.

Model: $I_d = \varepsilon_0 A\, dE/dt = \varepsilon_0 A\, E_0\, (2\pi f) \cos(2\pi f t)$, $E_0 = 10^5$ V/m, t = 0–1 ms.



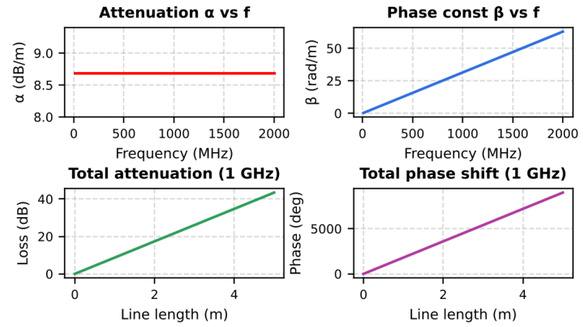
Result: Peak I_d = 55.6 μA at f = 1 kHz, A = 0.01 m². Doubling f (2 kHz) or doubling A (0.02 m²) both give 111.3 μA, while the E-field amplitude stays at 10^5 V/m.

Inference: $I_d \propto f$ and $I_d \propto A$, and it leads E by 90° (cosine vs sine). Plate area changes only the current, not the field.

3 Lossy Transmission Line: α and β — CO3 · BL4

Aim: Model a lossy transmission line from its per-unit-length R, L, G, C and study how frequency and line length affect attenuation and phase shift.

Model: $\gamma = \alpha + j\beta = \sqrt{[(R + j\omega L)(G + j\omega C)]}$; R = 50 Ω/m, L = 250 nH/m, G = 0.02 S/m, C = 100 pF/m, f = 10 MHz–2 GHz.



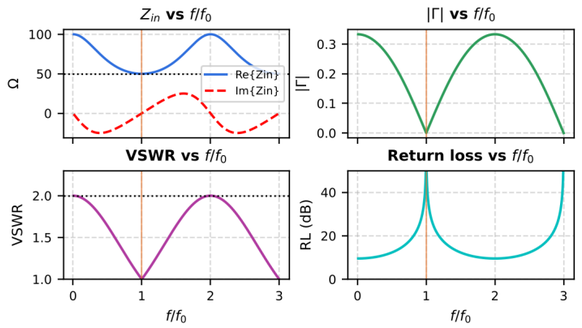
Result: At 1 GHz: α = 1 Np/m = 8.686 dB/m, β = 31.42 rad/m. Over 5 m: 43.4 dB total loss and 9000° phase shift.

Inference: Here R/L = G/C (= 2 × 10^8 s^{-1}), so the line is distortionless: α = √(RG) is constant with frequency and β = ω√(LC) rises linearly. Total loss and phase both grow linearly with length; Z_0 = √(L/C) = 50 Ω.

4 Quarter-Wave Transformer: 50 Ω to 100 Ω — CO4 · BL4

Aim: Design a quarter-wave transformer to match a 50 Ω line to a 100 Ω resistive load and examine the matching performance across frequency.

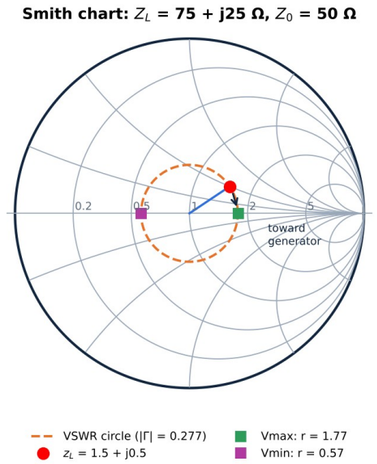
Model: $Z_T = \sqrt{(Z_0 Z_L)} = 70.71\ \Omega$; $\theta = (\pi/2)(f/f_0)$; $Z_{in} = Z_T(Z_L + jZ_T \tan\theta)/(Z_T + jZ_L \tan\theta)$



Result: f = f_0: Z_{in} = 50 Ω, |Γ| = 0, VSWR = 1.0 (perfect match). f = 0.9 / 1.1 f_0: |Γ| = 0.055, VSWR = 1.12. f = 0.5 / 1.5 f_0: |Γ| = 0.243, VSWR = 1.64. f = 2f_0 (λ/2 line): Z_{in} = Z_L = 100 Ω, |Γ| = 0.333, VSWR = 2.0.

Inference: The match is perfect only at f_0 (and odd multiples); away from f_0, Z_{in} turns complex and VSWR rises, so the λ/4 transformer is a narrow-band matching network.

5 Smith Chart: 75 + j25 Ω Load on a 50 Ω Line at 1 GHz — CO5 · BL4



Aim: Locate a 75 + j25 Ω load on the Smith chart of a 50 Ω line at 1 GHz and find Γ, VSWR, return loss and the positions of the voltage maximum and minimum.

Model: $z_L = Z_L/Z_0$; $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$; $VSWR = (1 + |\Gamma|)/(1 - |\Gamma|)$; $RL = -20 \log_{10}|\Gamma|$; λ = c/f = 30 cm (air line).

Results

Normalised load z_L = 1.5 + j0.5 (upper half → inductive)
Reflection coefficient Γ = 0.231 + j0.154 = 0.277 ∠ 33.7°
VSWR = 1.77 · Return loss = 11.14 dB · Mismatch loss = 0.35 dB
First V_{max}: 0.047λ = 1.40 cm from load (Z_{max} = 88.4 Ω)
First V_{min}: 0.297λ = 8.90 cm from load (Z_{min} = 28.3 Ω)

Inference: The load is a moderate inductive mismatch (≈ 7.7 % of power reflected). Moving toward the generator traces the VSWR circle; at the V_{min} point a λ/4 transformer of Z_T = √(50 × 28.3) ≈ 37.6 Ω would give a perfect match.